Yan Wang

(+1) 203 676 8734 yan.wang.yw2285@yale.edu | wangy05@whu.edu.cn YanWang-yale.github.io

EDUCATION

Wuhan University Wuhan, China

- B.S. in Biological Science; **GPA:** 3.91/4.00; **Major GPA:** 3.97/4.00; **Ranked #1** in the major Sep 2023-Jun 2027
- **Hongyi Honors Class:** Most Selective Program Integrating Interdisciplinary Study & Advanced Research
- **Academic Honors**: First-Class Outstanding Student Scholarship (top 5% of cohort, three consecutive years)
- **Senior Thesis: Yale School of Medicine** Jul 2026-May 2027

SCHOLARLY OUTPUTS

Manuscripts (*Corresponding author)

Wang, Y., Miller, L., Estrada, M., & Li, D.* Mitochondrial dysfunction and α-synuclein pathology in Parkinson's disease: A bidirectional pathogenic loop and therapeutic opportunities. *Frontiers in Molecular Neuroscience* (under review).

Wang, Y., Lin, Z., Miller, L., Estrada, M., Xian, J., Li, D.*, & IFGC. *PM20D1-AS1* links Parkinson's disease-associated genetic variation to *PM20D1* repression and α-synuclein regulation. (Manuscript completed; submission pending provisional patent filing).

Conference Abstracts (*Corresponding author)

Wang, Y., Zhang, J., Huang, X., Ruan, Y., & Yu, Y.* Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER-AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*. Abstract accepted for presentation at *Neuroscience 2026, Society for Neuroscience.*

RESEARCH EXPERIENCE

Visiting Undergraduate Researcher | Prof. Clemens Scherzer; Dr. Dunhui Li **Jul 2026-Present**

Adams Center for Parkinson's Disease Research, Yale School of Medicine New Haven, CT

Project: Functional Dissection of the PM20D1 Locus in Parkinson's Disease

- Identified a PD-risk SNP that strengthens the first splice donor of *PM20D1-AS1*, a Parkinson's-associated long noncoding RNA, promoting its maturation.
- Implicated *PM20D1-AS1* in cis-repression of the neighboring neuroprotective gene *PM20D1* using eQTL, ChAR-seq promoter contacts, and subcellular fractionation.
- Demonstrated that ASO knockdown increased PM20D1 protein and reduced PD-associated phosphorylated α-synuclein in human midbrain organoids.
- Bulk RNA-seq revealed restored synaptic/lipid-metabolism pathways and reduced SNCA/SCD expression in iPSC-derived cortical neurons following knockdown.
- Conducting ChIRP and RNAscope–immunofluorescence assays to investigate molecular interactions.

Undergraduate Researcher; Project I Lead | Prof. Yanxun Yu; Prof. Youngnam Jin **Sep 2024-Jul 2026**

Medical Research Institute, Wuhan University Wuhan, China

Project I: INS-1–Dependent Neuropeptidergic Tuning of Sensory Valence in C. elegans

- Elucidated the mechanism by which loss of the insulin-like neuropeptide INS-1 reverses high-salt ethanol avoidance into attraction in *C. elegans*.
- Mapped a distributed network of INS-1-releasing neurons in ethanol chemotaxis, identifying ASEL as a major source through neuron-specific knockout, rescue, and ablation.
- Identified an INS-1-dependent ASER sensory neuron-AIY interneuron circuit in which modulation of ASER ethanol sensitivity maintains downstream AIY inhibition to drive avoidance.
- Validated this circuit model through ASEL-specific INS-1 rescue at a major release site, post-rescue calcium imaging, and ethanol dose-response behavioral assays.
- Proposed NTR-1 as a candidate noncanonical INS-1 receptor by integrating pathway mutant analysis, *ntr-1* phenotyping, molecular docking, and MD simulations.

Project II: Performance Optimization of the POST-IT Proximity Tagging System

- Co-led optimization of POST-IT, a proximity-labeling platform using PafA, a bacterial protein-tagging enzyme, for small-molecule target identification.
- Developed a yeast-surface-displayed Pep1-MATH selection assay linking PafA catalytic activity to FACS enrichment.
- Conducted directed evolution of PafA using a ~10^6-variant error-prone PCR library and NGS-guided mutation tracking under progressively shorter labeling times and lower substrate concentrations.
- Combinatorially optimized selected PafA mutations to obtain a best-performing nine-mutation variant (9mt), reducing labeling from 18 h at 40 μM to 2 h at 10 μM and increasing threshold-positive cells from 0.5% to 27.2%.

Wet Lab Leader; Project Prototype Designer | Advisor: Prof. Lei Bai **Nov 2024-Oct 2025**
College of Life Sciences, Wuhan University Wuhan, China

Project: LOGIC | ***Gold Medal, 2025 International Genetically Engineered Machine (iGEM) Competition***

- Originated and designed LOGIC, a distributed biocomputing architecture using quorum-sensing (QS) signals to perform logic across bacterial colonies, overcoming single-cell circuit constraints.
- Screened QS systems to identify two sets of three mutually orthogonal signaling channels for independent intercolony communication.
- Engineered high-pass and band-pass response circuits and integrated them with QS modules into a reusable biocomputing toolkit.
- Demonstrated AND/OR/XOR, cascaded logic, spacing-dependent reconfiguration, and a prototype adder; trained a time – distance – concentration model to predict colony responses.
- Developed optogenetic reset modules using split-VVD-AiiA for AHL degradation and LOVdeg-tagged outputs.

TECHNICAL SKILLS

Models & Cell Culture: *E. coli*, *S. cerevisiae*, *C. elegans*, human iPSCs, iPSC-derived neurons, midbrain organoids.

Molecular & Biochemical Methods: Molecular cloning; RNA techniques (e.g., qPCR, RNAscope, ChIRP); protein expression and purification; Western blot; co-immunoprecipitation; immunostaining; subcellular fractionation; FACS.

Neuroscience & Functional Techniques: *C. elegans:* neuronal calcium imaging, optogenetic manipulation, neuron-specific genetic manipulation, microinjection, genetic crossing, and behavioral phenotyping; confocal microscopy.

Computational Genomics & Structural Biology: Bulk/snRNA-seq analysis; differential expression, alternative splicing, and eQTL analysis; RNA interaction modeling; molecular docking; molecular dynamics (MD) simulations.

HONORS & AWARDS

Grand Prize, Baosteel Outstanding Student (¥20,000), sole WHU nominee; 25 recipients nationwide; Nov 2026
1st Place, Hongyi Honors College's Annual Top Ten Student Innovators Award (¥10,000) Sep 2026
Qualified for Westlake University Rising Star Award (¥30,000; top 3%) Aug 2026
Liu Daoyu Creative Learning Scholarship (¥8,000), awarded to top 24 university-wide students Jan 2026
Gold Medal, International Genetically Engineered Machine Competition (**iGEM**) Nov 2025
Gold Prizes (Regular, Synthetic Cell, Protein Design & Bacterial Game Tracks) | **SynBio Challenges** 2025-2026
Top Ten Science Advances Award, Hongyi Honors College (¥10,000) Oct 2025
Outstanding Student Scholarship (top 5% cohort), Outstanding Student & Outstanding Student Leader 2024-2026
First-Class Outstanding Freshman Scholarship Sep 2023

ACADEMIC TRAINING & LEADERSHIP

Teaching Assistant, Biochemistry II, Wuhan University Nov 2025-Jan 2026
Team Advisor, WHU-China iGEM Team Nov 2025-Present
Participant, Neuroscience & Cognitive Science Summer Training Program, **Peking University** Jul 2026
Participant, "Frontiers of Life Sciences" International Summer School, **Westlake University** Jul-Aug 2025
Delegate, National "Top-Tier Talent 2.0" Biology Forum, **Tsinghua University** May 2025